

The AI Threat Landscape: What's Real vs Hype

The reality landscape (no hype)

Already deployed (defense):

- ML-based NDR/NDR (anomaly detection on flows, endpoints)
- UEBA (user & entity behavior analytics)
- Auto-triage for alerts (SOAR + ML classifiers)
- AI in CVE/prioritization triage

Already deployed (offense):

- AI-generated phishing content (human-like language, personalization)
- Automated recon at scale (osint-bots, web scraping)
- Polymorphic malware concepts (AI-driven mutation studies)

Emerging but not everywhere:

- Autonomous lateral movement (research-grade, fragile)
- AI malware agents (long distance from 'autonomous APT' – more agent-assisted)
- Quantum factoring (serious research, not yet threat-real at crypto scale)

The hype filter

Claim	Reality
"AI will autonomously compromise any org"	Fragile; needs infra, context, permissions. Not the near-term threat
"AI will replace SOC analysts"	AI augments triage; analysts provide context/judgment

Claim	Reality
"AI makes phishing impossible to detect"	Content-aware defenses (behavioral, simulation) still work; deepfakes are the real worry
"Quantum will break all encryption"	QC breaks some schemes (RSA/ECC); post-quantum crypto already deploying

The defender's advantage

ML is better at *detecting patterns* than producing novel, persistent offensive chains

AI-as-a-tool for blue (UEBA, triage) scales well

The real challenge: attacker content gets *more human* (phishing) – detection needs to follow

What you must learn

The actual capabilities (not vendor promises)

Where ML is *good* (anomaly at scale, triage) vs *bad* (novelty, context)

What you *will* defend against in the next 2 years (not the theoretical 20-year threat)

Hype tells you what to fear; reality tells you what to prepare.

Defenders who study the actual tech edge out those listening to the marketing podcast.

[← Module start](#)

[Deepfakes, Voice Cloning & AI ...](#)